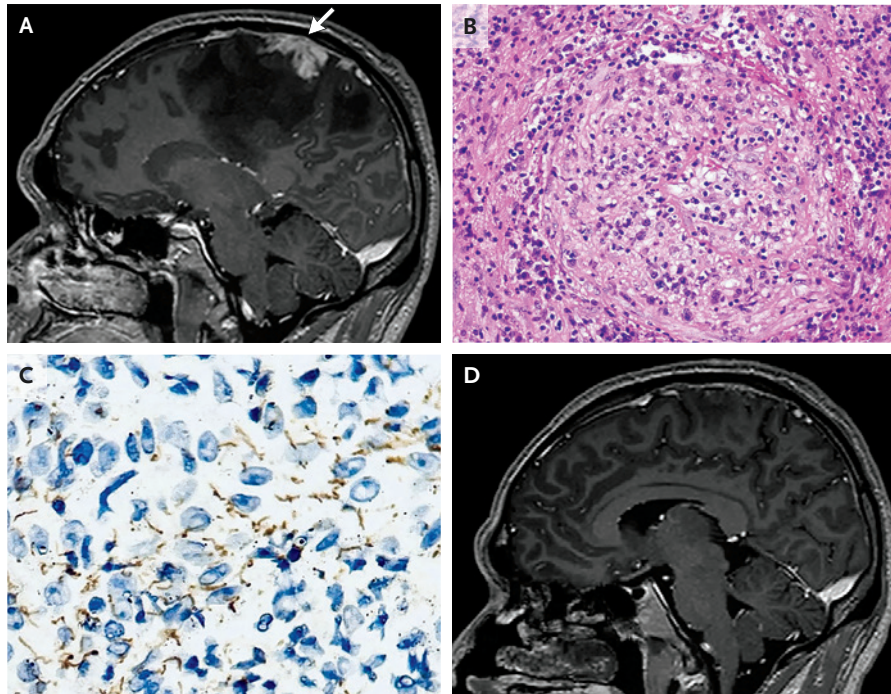


IMAGES IN CLINICAL MEDICINE

Cerebral Syphilitic Gumma



A 50-YEAR-OLD MAN PRESENTED TO THE EMERGENCY DEPARTMENT WITH a 1-month history of headache and worsening weakness of the left arm and leg. Physical examination was notable for the patient's inability to raise his left arm against gravity or lift his left leg against resistance. Magnetic resonance imaging of the brain after the administration of contrast material showed an enhancing lesion with extensive surrounding edema adjacent to the cerebral falx in the right frontoparietal lobe (Panel A, arrow; T1-weighted, sagittal view). A serum *Treponema pallidum* particle agglutination (TPPA) assay was positive, and the titer was 1:16 on a toluidine red unheated serum test (TRUST). Testing for human immunodeficiency virus was negative. An analysis of the cerebrospinal fluid showed a lymphocytic pleocytosis with a positive TPPA result. Histopathological analysis of a biopsy sample of the brain lesion showed abundant plasma-cell infiltration with arteriolitis (Panel B) and spirochetes on immunohistochemical staining (Panel C). A diagnosis of cerebral syphilitic gumma — a rare manifestation of long-standing syphilis — was made. Treatment with a 2-week course of intravenous penicillin was given. Seven weeks after the completion of treatment, the weakness of the left side had abated, the brain lesion had substantially decreased in size (Panel D), and the TRUST titer had decreased to 1:2.

DOI: 10.1056/NEJMicm2518333

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This article was published on May 9, 2026, at NEJM.org.